ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

# ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

## PAPER 1 · PRACTICE A · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central NC(V) Level 4 Electronics and Digital Concepts for Robotics paper 1 · practice a.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and examples.

## SECTION A

## QUESTION 1

**1.1** D ✓
**1.2** B ✓
**1.3** C ✓
**1.4** A ✓
**1.5** B ✓
**1.6** D ✓
**1.7** A ✓
**1.8** C ✓
**1.9** C ✓
**1.10** A ✓

(10 × 1) [10]

## QUESTION 2

**2.1** False ✓
**2.2** True ✓
**2.3** True ✓
**2.4** True ✓
**2.5** True ✓
**2.6** False ✓
**2.7** False ✓
**2.8** True ✓
**2.9** True ✓
**2.10** False ✓

(10 × 1) [10]

## QUESTION 3

**3.1** F ✓
**3.2** C ✓
**3.3** J ✓
**3.4** E ✓
**3.5** H ✓
**3.6** D ✓

**3.7** G ✓

**3.8** A ✓

**3.9** I ✓

**3.10** B ✓

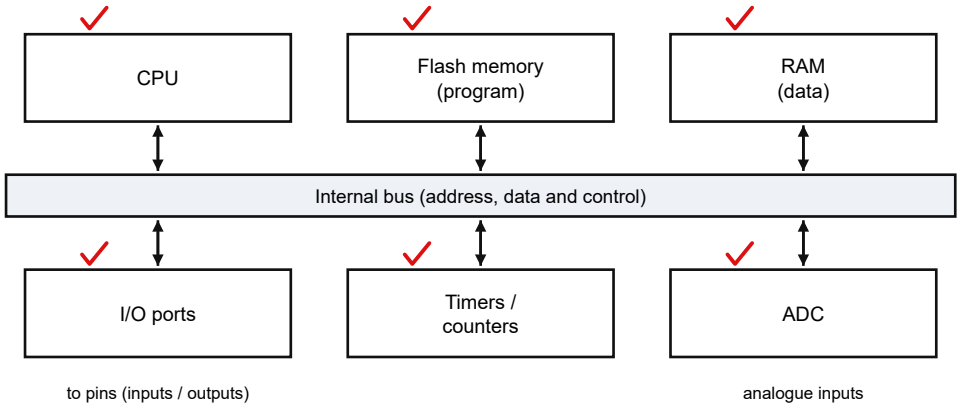
(10 × 1) [10]

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 4

**4.1**




CPU ✓; flash (program) memory ✓; RAM ✓; I/O ports ✓; timers ✓ and ADC ✓, all connected by an internal bus (address/data/control)

(6)

**4.2** A microcontroller has the CPU, memory and I/O on one chip, designed to control one dedicated task ✓, e.g. ATmega328P (Arduino Uno), PIC, ESP32 ✓; a microprocessor is only the CPU and needs external memory and I/O, used in general-purpose computers ✓, e.g. Intel Core i5, the processor of a Raspberry Pi / ARM Cortex-A ✓. (4)

**4.3** UART: 2 wires (TX, RX) ✓; slow to medium (e.g. 9 600–115 200 baud) ✓; GPS, Bluetooth module, PC serial monitor ✓. I²C: 2 wires (SDA, SCL) ✓; medium (100–400 kHz) ✓; IMU, OLED display, real-time clock ✓. SPI: 4 wires (MOSI, MISO, SCK, SS) ✓; fast (several MHz) ✓; SD card, TFT display, radio module ✓ (9)

**4.4** With polling, the program must keep checking the pin and can miss pulses while it is busy ✓; an interrupt responds immediately to every pulse ✓, so counting is accurate and the main program can do other work ✓. (3)

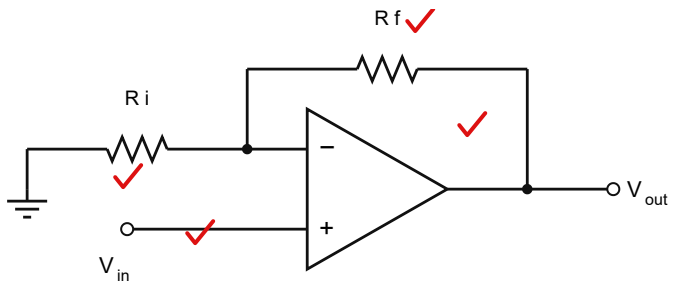
**[22]**

## QUESTION 5

**5.1.1** $A_v = 3{,}3 \div 0{,}3$ ✓ $= 11$ ✓ (2)

**5.1.2** $A_v = 1 + R_f/R_i \rightarrow 11 = 1 + R_f/1\ k\Omega$ ✓; $R_f = 10 \times 1\ k\Omega$ ✓ $= 10\ k\Omega$ ✓ (3)

**5.1.3**



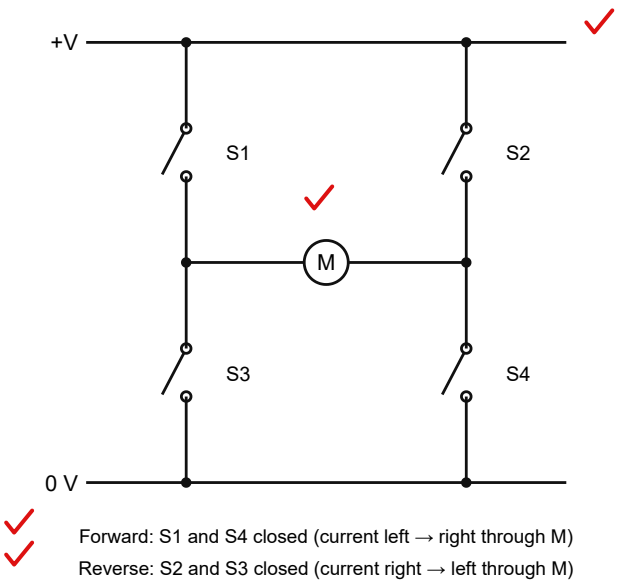

Op-amp triangle ✓; input to the non-inverting (+) terminal ✓; $R_f$ from output to the inverting (−) terminal ✓; $R_i$ from the inverting terminal to 0 V ✓ (4)

**5.2** $V_{out} = -(R_f/R_i) \times V_{in}$ ✓ $= -(100/20) \times 0{,}5$ ✓ $= -2{,}5$ V ✓ (3)

**5.3** $1000\ 0000 = 128$ ✓; $V = 128 \times 5 \div 256$ ✓ $= 2{,}5$ V ✓ (3)

**[15]**

## QUESTION 6




**Example of a correct layout for 6.1 (other neat, correct answers get full marks)**

**6.1** Diagram: four switches S1–S4 arranged in an H, with the motor across the middle ✓✓, supply at the top and 0 V at the bottom ✓. Forward: S1 (top left) and S4 (bottom right) closed ✓✓; reverse: S2 (top right) and S3 (bottom left) closed ✓✓ (accept any consistent labelling) (7)

**6.2** It would short-circuit the supply (shoot-through) ✓, causing a very high current that can destroy the transistors or battery ✓; driver ICs have built-in logic / dead time that prevents both switches on one side from being on together ✓. (3)

**6.3** When the motor stalls or starts, it can draw more current than the driver can supply, so the driver may overheat or be damaged ✓. Any TWO ✓✓: use a driver rated above the stall current (e.g. 3 A or more); add current limiting / fuse; use a heat sink and cooling; avoid stalling with gearing; use soft starting with PWM ramps (3)

**[13]**

## QUESTION 7

**7.1.1** $P = (V_{in} - V_{out}) \times I$ ✓ $= (9 - 5) \times 0{,}5$ ✓ $= 2$ W ✓ (3)

**7.1.2** 2 W makes the regulator very hot, and it may shut down or be damaged ✓; a heat sink spreads the heat ✓; a switching (buck) converter is more efficient (about 85–95%) ✓. (3)

**7.2** $t = \text{capacity} \div \text{current} = 2\,200 \div 550$ ✓ $= 4$ h ✓; reasons (any TWO) ✓✓: current peaks when motors start or climb; battery should not be fully discharged; capacity drops with age and temperature; voltage sag (4)

**7.3** Any FIVE ✓✓✓✓✓: use a proper LiPo balance charger; never leave batteries charging unattended; charge on a fire-proof surface or in a LiPo bag; do not charge damaged or swollen batteries; do not over-discharge (stop at about 3,0–3,3 V per cell); store at storage voltage (about 3,8 V per cell) in a cool place; avoid short circuits / cover terminals (5)

**7.4** A common ground gives all signals the same 0 V reference ✓, so the microcontroller's control signals are read correctly by the motor driver ✓. Large, changing motor currents cause voltage drops and electrical noise ✓, which can reset the microcontroller or corrupt sensor readings ✓; so the grounds are joined at one point (star ground) and motor wiring is kept separate / decoupling capacitors are used ✓. (5)

**[20]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**